

Sorting, Not Calibration: A Price-Blind Comparison of Language Models and Prediction Markets

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Abstract—We compare eight language models with prediction-market prices on Polymarket and Kalshi. The benchmark has 10,536 market moments, and the headline results use the 4,865 held-out primary rows. The models never see a price or order book, and most rows carry no news, so they work with less information than the market. The market beats every model. The gap is in sorting, not calibration: the models separate likely from unlikely outcomes less well than the crowd does. Resolution accounts for 79 to 99 per cent of each model’s shortfall, and no monotone rescaling recovers more than about a fifth of it, or two fifths on the non-sports rows, where the gap is widest. News improves sorting for nearly every model, significantly only on the window whose news carries a leakage caveat, and never to the market’s level. The models are also wrong together. On rows the market got right, two models land on the same wrong side 2.7 to 4.3 times as often as independent forecasters would. Simulated forecasters that share only each model’s fitted line to the price reach 1.4 to 2.2. Adding each row’s features or its event still leaves them below the real rate. News reduces this agreement only slightly, and we cannot tell missing information apart from shared training or the shared prompt. No combination of the models, equal-weighted or learned, matches the market. Returns are no worse in thin markets, though the accuracy gap narrows in deeper order books, clearly on one window and less robustly on another. These results are exploratory. We release the benchmark, model outputs and code, except one window’s market data, which cannot be redistributed.

1 Introduction

A prediction market turns a question about the future into a contract that pays one dollar if the answer is Yes and nothing if it is No. Its trading price is a probability with money behind it, and decades of work have found those prices hard to beat [1]. Recent studies report that large language models now forecast about as well as human crowds [2], beat the median forecaster on public tournaments [3], and, in one recent workshop paper on prediction markets, match the price on accuracy while earning higher returns [4]. If that is right, it is a notable claim about machine judgment. It is also hard to test, because three things have to be true of the comparison.

The market has to be trading. On both venues a fresh listing carries a one-cent bid against a ninety-nine-cent ask, and the midpoint is fifty cents whatever the event’s real chance. Every row of the two live-book windows has a live book at the forecast moment: quotes on both sides, with a spread of ten cents or less. The third, PolyBench’s published week, was drawn without that rule, and we rest no claim on it alone.

The model has to form its own number. If a model is shown the market price, whatever it returns is contaminated by the number it was handed. Language models move markedly toward a crowd estimate when shown one [2]. No model in this study sees the price, the best bid, the best ask, the depth or the book; it sees the question, the resolution rules, the scheduled close and a frozen news snapshot.

The model must not already know the answer. Restricting retrieval by date does not prevent leakage: one audit found major post-cutoff information in 71 and 81 per cent of date-filtered pages, with the answer revealed outright for 41 and 55 per cent [5]. A model forecasts a row only if its exact published version was released before that row’s outcome could be known, the news was captured before the forecast moment, and no model performs a live web search.

With those three conditions in place, the market is more accurate than every model in every depth bucket on the two live-book windows. The gap is resolution rather than calibration: no monotone rescaling recovers more than about a fifth overall. The losses concentrate where a model contradicts the crowd (Section 6.3). The models also err together: on rows the market got right, two models land on the same wrong side 2.7 to 4.3 times as often as independent forecasters would. Controlling for what they were jointly given leaves that rate well above what those inputs alone produce, and adding news lowers their agreement only slightly; information none of them had, shared training and a shared prompt remain. No combination of the models reaches the market (Section 6.4).

We contribute a replay benchmark of 10,536 market moments on two venues, a price-blind evaluation harness (the code that prompts each model and records its answer) run over eight release-date-eligible models, the finding that the accuracy gap lies on the resolution axis, checked against recalibration and learned combination, and a correlated-error measure tested against shared shrinkage, the observable features of a row and the event itself.

2 Related Work

Language models as forecasters. An ensemble of twelve models matches a human crowd on tournament questions [2], and frontier models now score better than the median Metaculus forecaster but behind assembled experts [3]. Pairing thousands of tournament questions with a date-organised news corpus left model accuracy far below a human expert baseline, though retrieval helped [6]. On a rolling benchmark, expert forecasters

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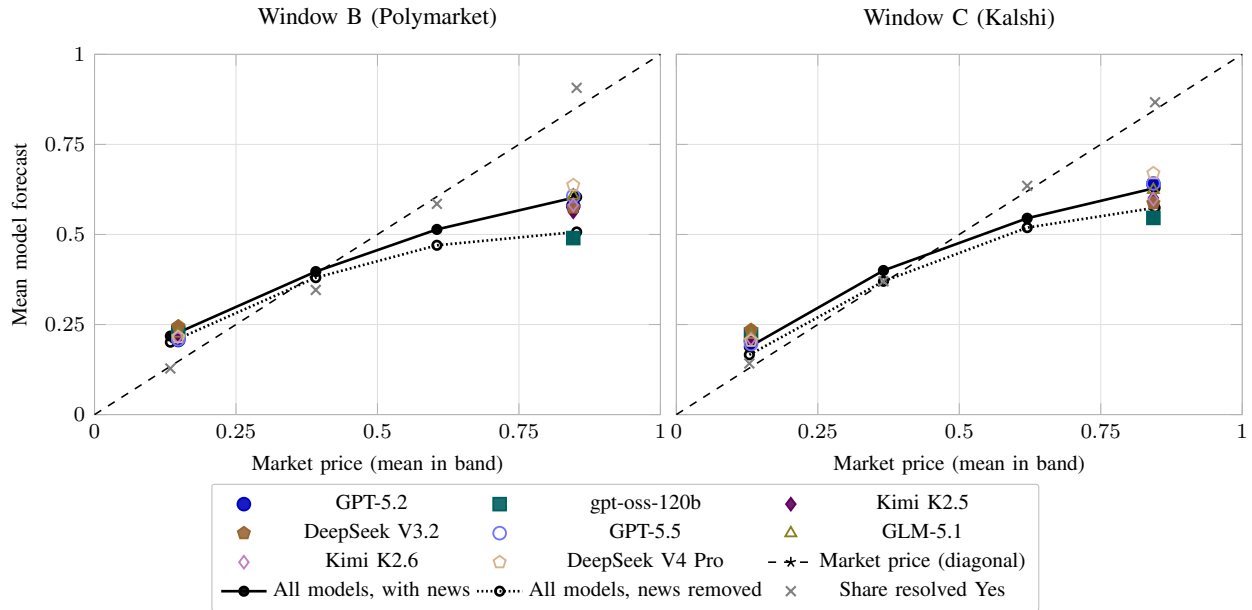


Fig. 1: No model sees the market price, yet the forecasts flatten against it. Black lines are all models pooled on news-bearing rows, with (solid) and without (dotted) the news, in four price bands. Grey crosses are the actual Yes share on those rows. Coloured markers are each model’s mean forecast in the lowest and highest band, over all primary rows. Test split; pooled lines use news-bearing rows, per-model markers use every row. Top-band Yes share: 91 and 87 per cent on news-bearing rows, 86 and 84 on all.

still beat the best model at $p < 0.001$ [7]. A retrieval system scored 0.179 in Brier against the crowd’s 0.149, passing it only on selected subsets [8].

Models trading live on Polymarket and Kalshi returned -16 to -31 per cent on Kalshi over 57 days [9]. A recent workshop paper [4] reports a model whose accuracy is indistinguishable from the price. Our design differs on all three counts of Section 1. Profit need not track accuracy: a forecaster less accurate than the price can still profit by disagreeing with it [10].

What the model is shown. PolyBench [11] covers 38,666 binary Polymarket markets in February 2026, but its prompt injects order-book spreads and midpoint prices. So does Prophet Arena [12], whose frontier models match the market in Brier score and beat it on calibration and return. Window A of our benchmark is PolyBench’s published data: we reuse its market rows and news, not its protocol. Every model here is asked with no price, bid, ask or book in the prompt, as in Hindcast [13], which replays 216 Polymarket markets with small open-weight models but does not compare their Brier score with the price’s.

What the price is. Thin markets carry known distortions, such as the favourite-longshot bias, under which cheap contracts are overbet [14]. Our models do buy longshots when they lean against the crowd, and lose on them through fees and mispricing in proportions that vary by window (Section 6.3).

Leakage. Date-filtered retrieval does not prevent leakage [5], so we use no retrieval at forecast time and admit a model by its published release date (Section 4).

Scoring. We score with the Brier score [15] and split it with Murphy’s partition [16]. Binning is not harmless [17],

so we repeat the decomposition without bins using isotonic regression [18]; the attribution to resolution is unchanged (Section 6.2). We also refit each model with a Platt sigmoid [19] on held-out data.

Correlated errors. Across more than 350 models, two models that are both wrong give the same wrong answer 60 per cent of the time, and the correlation is strongest among the largest [20]. Three models on 568 forecasting questions show a mean pairwise error correlation of 0.77 [21]. We correlate departures from the market price instead, because raw error correlation overstates agreement when forecasts move little relative to outcomes, and provide a shrinkage simulation as the null (Section 6.4). When decision makers adopt the same rule, errors become structural [22], and the right amount of extremizing falls to nothing as forecasters’ information overlaps [23].

3 Benchmark

A row is a market-moment. One row is one market at one instant. The model-visible part is the question, event title, resolution rules, scheduled close and a frozen news snapshot. The analysis-only part (order book, price, outcome, price path, fee schedule) is never shown to any model; the separation is enforced in the harness.

Three windows, never pooled. Window A is PolyBench’s February 2026 week on Polymarket [11], with every model asked under our protocol. Window C is Kalshi over four weeks in May–June 2026, and Window B is Polymarket over 26 days in July–August 2026 (Table I). The letters are labels, not calendar order.

Sampling. Each market enters at most once, on the first day its 18:00 UTC order book is live: quotes on both sides, spread ten cents or less. Each day contributes 120 non-ladder rows (24 from each of five per-day depth groups, not the analysis buckets of Table I) plus 20 ladder rows; one Window B day had only 99 eligible non-ladder markets. Ladders (*Bitcoin above \$70k, \$72k, \$74k*) stay out of the primary set and are scored on their own in Section 6.1. Window A was drawn without the live-book rule; we use its rows as PolyBench published them.

The market’s probability. We take the crowd’s probability to be the midpoint of the best bid and best ask at the forecast moment. Window B and Window C store that midpoint directly. On Window A the stored price is a stale sweep, so we recompute the midpoint from the book for consistency across windows.

News. For Window B and Window C we built a frozen snapshot from the Common Crawl news archive, captured strictly before the forecast moment; rows whose news carries a late dateline are dropped. Window A keeps the articles PolyBench supplied, stamped as fetched at the forecast moment; 159 of its rows carry a dateline at least two days after that stamp, so those articles were fetched later, and the rows are dropped. About seven in ten Window B and Window C rows have no news at all, so results distinguish two arms: rows with news and all rows. The attachment rules are in Appendix D.

The primary analysis set. Headline numbers use a filtered subset. Ladder rows go because their prices are formulas. Rows whose end date appears to have moved go because the model may have answered a different question from the one that settled; detection relies on internal inconsistency (slug date versus game start, or scheduled close versus a date in the question text), so a clean edit that left no trace would pass (Section 7). This fires on 253 Window B rows and none elsewhere. Late-dateline rows go for the leakage reason above (159 of 2,997 on Window A). Rows priced outside 0.05 to 0.95 go because near the boundaries one surprise swamps the comparison. What remains is 1,550, 1,865 and 2,669 rows.

Split. Each window’s primary set is divided 20 per cent calibration and 80 per cent test, drawn within each window-by-depth-bucket-by-sports cell so the two halves match in composition, under a fixed seed. The recalibration control of Section 5 is fitted on the calibration split, so it never sees the rows it is judged on.

Depth. Depth is the dollars resting on the five best prices on each side of the Yes book at the forecast moment. Each window’s primary set is cut into five equal-count buckets (Table I), formed within a window, so bucket 1 always means thin relative to that window.

Fees. Trading results charge each venue’s real taker-fee schedule per row; both schedules peak at even money, which is where most disputed trades sit. The exact formulas are in Appendix C.

4 Models and Evaluation Harness

Roster and eligibility. Eight models are scored. A model may forecast a row only if its exact published version was

released before the row’s outcome could be known; in practice every model was released before its first window opened. The four models published between August 2025 and January 2026 run on all three windows. The four published between 7 and 24 April 2026 are ineligible for Window A (February 2026) and eligible for Window C (May to June) and Window B (July to August). The rule is a property of dates rather than of a leakage test that might fail silently. The full roster is in Appendix V.

Hosts and settings. GPT-5.2 and GPT-5.5 are served by their developer; OpenAI returned dated snapshots on every call, confirming the checkpoint did not change. Every open-weight model is pinned to a single host with fallback routing off, and the served model string is checked on every reply. Settings are fixed: temperature 0 where the host accepts it, one sample per row, 4,000-token reply limit. Thinking is a reasoning-effort level of *medium* on the OpenAI models and gpt-oss-120b, and a 1,000-token budget on the others; the asymmetry cannot be removed (Section 7).

Prompt. Every model receives the same template (Appendix W), requiring a final `PROBABILITY:` line. It contains no price, bid, ask, depth or book. All scored calls use this one template.

Forced answers. An unparseable reply is re-asked once with a stricter instruction; a second failure is recorded as failed. Across all scored runs 79 rows failed, all from four models, 42 inside the primary set. On the test split, filling every failure with 0.5 moves no Brier edge by more than 0.0011; filling each with the worst possible answer moves none by more than 0.0077, always in the direction that makes the models look worse. The re-ask was needed on 4 to 14 per cent of rows for Kimi K2.5, about 5 per cent for DeepSeek V4 Pro and at most 3 per cent for every other model.

Leak checks at call time. Two checks run before every call. The first rejects a row whose news capture time is not strictly before its forecast moment. The second scans for a dateline two or more days after the forecast moment, besides the benchmark’s own flag, which is anchored on the capture time (Appendix D). A row that would leak is never sent.

Memory probe. Every news-bearing row is asked again with the news removed. The pair measures what the news was worth and flags rows a model may have memorised; flag rates over whole probe runs are 0.8 to 2.3 per cent. Appendix X gives the rule and a validation exercise whose result is genuinely inconclusive (Section 7).

Scale and cost. The study is 97,465 model calls, costing about \$620.

5 Evaluation

Accuracy. We score with the Brier score [15], $BS(p) = n^{-1} \sum (p_i - y_i)^2$, and compare a model with the crowd by the *Brier edge* $\Delta = BS(q) - BS(p)$, so positive means the model was more accurate. We report the difference rather than two levels because the level depends on how hard a window’s questions happened to be.

TABLE I: The benchmark. *Drawn* is every row sampled; *in play* drops the 159 Window A late-dateline rows. The primary set drops ladders, 253 moved-end-date Window B rows, and prices outside 0.05–0.95; headline numbers use its test split. Depth cuts are the four boundaries between five equal-count buckets within each window.

Window	Venue	Dates	All rows		Primary set				Depth cuts (\$)
			Drawn	In play	Rows	Test split	With news	Sports	
A	Polymarket (PolyBench)	6–12 Feb 2026	2,997	2,838	1,550	1,248	1,434	82%	293 / 619 / 1,672 / 11,334
B	Polymarket	15 Jul–9 Aug 2026	3,619	3,619	1,865	1,480	563	73%	330 / 775 / 1,676 / 4,421
C	Kalshi	14 May–10 Jun 2026	3,920	3,920	2,669	2,137	752	81%	566 / 1,472 / 2,692 / 4,761

Splitting the score. We separate the Brier score with Murphy’s partition [16],

$$BS = \underbrace{REL}_{\text{calibration error}} - \underbrace{RES}_{\text{sorting}} + \underbrace{UNC}_{\text{fixed by outcomes}}, \quad (1)$$

computed over ten equal-width bins. Reliability measures whether stated probabilities are honest; resolution measures whether the forecaster separates events that happen from events that do not. Uncertainty depends only on the outcomes, so it cancels from the edge.

Recalibration control. If the gap were a calibration failure it could be repaired after the fact by rescaling. We test that directly, fitting the two-parameter Platt sigmoid [19] to each model’s probabilities on the calibration split and scoring the mapped probabilities on the test split. We report the fraction of the Brier gap it recovers.

Money. We simulate trading from the book at the forecast moment: if the model’s probability is above the best ask we buy Yes, if below the best bid we buy No, otherwise no trade. Each venue’s real taker fee is charged. Rows with a one-sided or crossed book, or a spread wider than 0.50, are never traded. Returns are per dollar staked.

Direction. A model *leans with the crowd* on a row when its probability sits further toward the side the market already favours than the market price does, and *against* the crowd when it sits on the far side of the price from that side. A model exactly at the price, or any model on a row the market prices at exactly 0.5, leans neither way, which is why the two shares do not sum to one. A model that says 0.60 when the market says 0.70 counts as leaning against the crowd, even though both favour Yes; the model is less confident, not contradicting.

News value. The value of news is the Brier score without it minus the Brier score with it, so positive means the news helped. A forecast moved *toward* the market when the with-news answer is closer to the price; movements smaller than 0.02 count as neither toward nor away.

Correlated errors. Raw error correlation overstates shared judgment when forecasts move little relative to outcomes, so the primary measure correlates *departures from the market price*, $p - q$; because two forecasters that shrink toward the same base rate agree on departures regardless, we fit each model’s own shrinkage and correlate the residuals. The null is simulated forecasters carrying each model’s fitted shrinkage but independent noise. *Wrong together* is the share of model pairs on the same incorrect side of even money, compared with the independent rate. Equations and construction are in Appendix F.

Exploratory status. The analyses in this paper, including the scripts of Appendix U, are exploratory.

Depth. We measure whether losses grow as the book thins by the thin-minus-deep return gap and the slope of return on the natural log of depth.

Uncertainty. All intervals are 95 per cent bootstrap ranges resampling *events* rather than rows, because markets within one event move together and resampling rows would give intervals that are too narrow. The main tables use 2,000 resamples; exploratory scripts use 1,000.

Which rows a number uses. Every result is computed on the test split or a named subset of it. The exceptions are Table I, which describes the benchmark, counts over whole runs (failures, probe flag rates), and the cross-fitted variants below. The calibration split is used to fit the recalibration control and the combination rules, which are then scored on the test split; the cross-fitted variants score every row because no held-out set remains (Figure 4). Unless a number is explicitly about news it is the all-rows arm.

Gates and quarantine. A calibration gate on model-window-bucket cells (Appendix R) and the memory-probe quarantine (Appendix X) apply only to the convergence test of Appendix S. Every other number, Table II included, uses all rows, flagged ones too.

6 Results

6.1 The market beats every model

Every one of the 80 points in Figure 2 lies below zero: on Window B and Window C the market was more accurate than every model in every depth bucket. The cells share rows across models, so the 80 describe the data rather than confirm it 80 times; the independent variation is across windows and buckets.

All 40 Window C cells are clear of zero; on Window B 31 of 40 are, and the nine that are not form a pattern: the deepest bucket for all eight models, plus GPT-5.5 in the thinnest. The market’s advantage is smallest in the deepest bucket, consistent with the depth pattern of Section 6.5.

The models do beat a naive forecaster. Measured as skill against a forecaster that always says the base rate, the market scores +0.199, +0.207 and +0.232 on the three windows, and the roster average scores +0.121, +0.085 and +0.072. As a counterfactual on the price, pulling the market toward the base rate along the roster’s average fitted line to it scores +0.158, +0.168 and +0.194, still better than every model; that pull alone reproduces 52, 32 and 24 per cent of the shortfall. The

slope below one, which we call shrinkage, need not mean the models hedge: a calibrated forecaster that knows less than the market falls back toward the base rate in the same way, and on Window B and Window C the recalibration fits of Section 6.2 point away from hedging (on Window A, where their slopes exceed one, toward it).

Whole-window gaps run from 0.0214 (GPT-5.5) to 0.0416 (gpt-oss-120b) on Window B and from 0.0272 to 0.0431 on Window C, every one clear of zero, as are all four on Window A (Table III). The spread between models is smaller than the distance from any of them to the market.

Scoring every non-ladder in-play row instead, moved-end-date rows and extreme prices included, gives Brier edges of -0.0257 to -0.0455 on Window B and -0.0281 to -0.0425 on Window C; ladder rows on their own are worse still on Window B, -0.055 to -0.067 , and similar on Window C, -0.024 to -0.037 . Every interval is below zero.

Window A is the one window where the picture is not uniform: the edge is negative in 15 of 20 cells and clear in 8. It is also the smallest window, the only one drawn without the live-book rule and the only one whose news came from live retrieval rather than a frozen archive. We report it throughout but rest no claim on it alone.

Part of the gap is information. News closes about a third of the sorting gap on Window A and about a sixth on Window B, changing resolution by -0.002 to $+0.012$ and raising it for 18 of the 20 model-window pairs. The effect is individually significant only on Window A, where three of four models have intervals above zero; on Window B and Window C every per-model interval includes zero, and one model on each (GLM-5.1, DeepSeek V3.2) goes slightly the other way. That Window A is the one window with a significant effect needs a caveat: it is also the only window whose news was retrieved at run time rather than from a frozen archive, the setting in which post-cutoff material has been found leaking through [5]. The gap never closes: with news in hand, every model still sorts worse than the market. What these numbers bound is the value of our news (at most three archived articles, on the minority of rows carrying any), not of information in general; a market watching lineups, injuries and closing lines has far more.

The gap is several times wider on non-sports rows, so the sports concentration does not produce it; the models capture a smaller share of what is available on non-sports than on sports (Appendix P). The size of the deficit depends strongly on forecast horizon, with every model agreeing on the worst band within a window, but which end of the horizon hurts reverses between Window B and Window C (Appendix T).

GPT-5.5, one of the two most recently released models, has the highest resolution in the study and is still short of the market with both intervals clear of zero; its same-day peer DeepSeek V4 Pro does not match it (Appendix I).

6.2 Sorting, not calibration

Splitting those Brier scores with Equation (1) locates the gap. Figure 3 plots calibration error against resolution: the

models cluster near the market horizontally and far beneath it vertically.

Their reliability runs from 0.0032 to 0.0102 on Window B and from 0.0023 to 0.0106 on Window C, against 0.0017 and 0.0006 for the market. The models are worse here too, but the quantities are small on both sides, and reliability is a minor part of the shortfall, not its source.

On the sorting axis the models are not close. Resolution runs from 0.0167 to 0.0306 on Window B and from 0.0194 to 0.0302 on Window C, against 0.050 and 0.056 for the market, and the market-minus-model difference is clear of zero for all sixteen model-window pairs ($+0.019$ to $+0.033$ on Window B, $+0.026$ to $+0.036$ on Window C). By isotonic regression, resolution accounts for 87 to 99 per cent of each model's Brier gap on Window A, 82 to 97 per cent on Window B and 79 to 95 per cent on Window C. The ten-bin decomposition agrees at 79 to 94 per cent on Window B and 76 to 94 per cent on Window C; the bin-free figure avoids the within-bin bias that equal-width bins introduce when the two sides' answers cluster differently. The pattern holds within both halves of the sample but is weaker on non-sports rows, where the gap per row is widest: there resolution carries 73 to 90 per cent of the shortfall on Window B and 59 to 89 per cent on Window C, and a cross-fitted rescaling recovers up to 30 and 42 per cent (Appendix P).

Recalibration is a second view of the same fact, not an independent test, since reliability is roughly what rescaling can recover (Figure 4). The Platt map recovers 2 to 16 per cent of the gap on Window B and 6 to 18 per cent on Window C; cross-fitted Platt recovers 1 to 16 per cent and 5 to 15 per cent; a monotone step function reaches at best 19 and 20 per cent. Four fifths or more is beyond any monotone rescaling, which cannot reorder forecasts, and it is the ordering that is wrong. On Window A the fitted maps have slopes above one and recover between -9 and $+2$ per cent. Redrawing the calibration split to keep every event on one side does not raise recovery (Appendix L).

The Platt slopes (of the recalibration map, not of the fit to the price) on Window B and Window C are all below one (0.28 to 0.73 on Window B, 0.41 to 0.84 on Window C), meaning the right correction is to pull each forecast *in*, toward the map's own centre. Yet Figure 1 shows the models answering 0.55 where the crowd says 0.85 and 85 per cent of such rows resolve Yes. Both are true, of different things: the models are overconfident in their own noisy ordering (their spread is wider than their ability to discriminate justifies) and underconfident relative to the market's better ordering. That reliability comes out small is not evidence of good calibration: it is a squared quantity set against a resolution gap several times larger, the worst-calibrated forecasts sit in the sparse tails, and ten bins hide disagreement inside a bin. Splitting each bin by whether the market priced the row above or below it raises the measured value three- to fivefold (Appendix M).

Figure 1 shows what this means in practice. A calibrated model's 0.55 is right on average, but the rows behind it can mix markets the crowd prices at 0.85 with markets it prices at

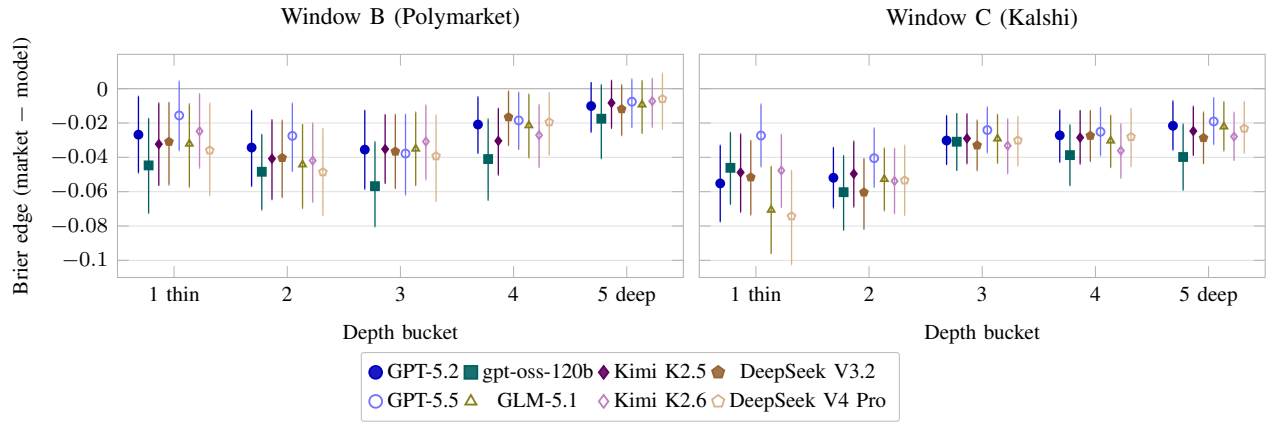


Fig. 2: The market’s Brier score minus each model’s, per depth bucket; points below zero mean the market was more accurate. All 80 points lie below zero and 71 of 80 ranges exclude it; models share rows within a bucket, so the 80 cells are not independent tests. Test split of the primary set, all rows; whole-window edges with intervals are in Table III.

0.40. The model’s 0.55 is honest about the average and silent about the difference, and it is the difference the Brier score pays for.

6.3 Losses sit where a model contradicts the crowd

If the models sort poorly, the damage should sit where a model disagrees with the crowd. Table II splits every row by direction. Rows leaning with the crowd cost little, at a Brier edge of $+0.005$ on Window A, -0.003 on Window B and -0.011 on Window C; rows leaning against sit far below, at -0.038 , -0.046 and -0.052 , and carry 110, 96 and 91 per cent of the net accuracy deficit (over 100 on Window A because there the with-the-crowd rows favour the models). That much follows from shrinkage alone: the against category is 53, 60 and 66 per cent of rows, and 66.8, 69.0 and 60.4 per cent of it is the model favouring the market’s outcome with less confidence. The loss sits where the model favours the other outcome: those rows are 18, 18 and 26 per cent of all rows, score -0.084 , -0.121 and -0.104 against the market, and carry about 80, 78 and 72 per cent of the net deficit (Appendix J).

In money the models systematically buy longshots [14]: against-the-crowd trades buy at 0.28 to 0.31 and win 27 to 29 per cent of the time, returning -0.172 per dollar on Window A and -0.124 on Window C. On Window B the money half is not conclusive: neither group’s interval excludes zero, though the accuracy half is clear on all three windows. Fees are about three quarters of the trading loss on Window B and two fifths on Window C; on Window A, where fees were close to nothing, the loss is mispricing (Appendix N).

Figure 1 shows where the disagreement lives. In the highest price band the crowd says 0.846 on Window B and 0.843 on Window C and about 85 per cent of those markets do resolve Yes; the models answer 0.490 to 0.637 and 0.546 to 0.670. Each model’s forecast fitted against the price has a slope on the price well below one (roster mean 0.57 on Window B; individual slopes 0.45 to 0.65), and its fixed point sits near the window’s base rate. The Yes side costs $+0.192$ Brier on Window A, $+0.082$ on Window B and $+0.075$ on Window C

more than the No side, the pooled interval above zero on each window (Figure 1).

News does not fix this. Scored directly, it is worth -0.001 to $+0.017$ in Brier (Appendix X). A toward/away split of the news effect appears large, but a random-direction placebo reproduces 99 per cent of it on Window B and 102 per cent on Window C, and overshoots it on Window A: because the price predicts the outcome, any move landing closer to it scores better whether or not it carried information.

6.4 Correlated errors

The models do not fail independently. Simulated forecasters that shrink as hard but share nothing else correlate at only 0.23 to 0.28 in departures from the price; the real departure correlations are 0.801 on Window A, 0.795 on Window B and 0.775 on Window C (Figure 6 in Appendix O). Removing each model’s own shrinkage and correlating what is left gives 0.750, 0.734 and 0.708, against a simulated value near zero (Figure 5).

That last comparison is weaker than it looks: the simulated forecasters are built with independent noise, so their residual correlation is near zero by construction. The comparison worth making is against explanations that do not require shared judgment. Every model saw the same question and the same news, so agreement might reflect a common brief. Adding the row’s observable features to the fit (sports flag, topic, horizon, price band, news indicator) barely moves it: 0.711, 0.726 and 0.695, a fall of about five per cent at most. Adding a per-event intercept, which absorbs everything the markets of one event share whether measured or not, gives 0.747, 0.662 and 0.694 on the 942, 475 and 511 rows that belong to events with more than one market. Those rows are not the full test split, so the comparison that matters is against themselves: price alone gives 0.785, 0.701 and 0.714 on the same rows, so the event intercept removes about 0.04, 0.04 and 0.02. Neither the observable characteristics nor the event itself accounts for this residual agreement, though neither would capture a shock specific to one market at the forecast moment.

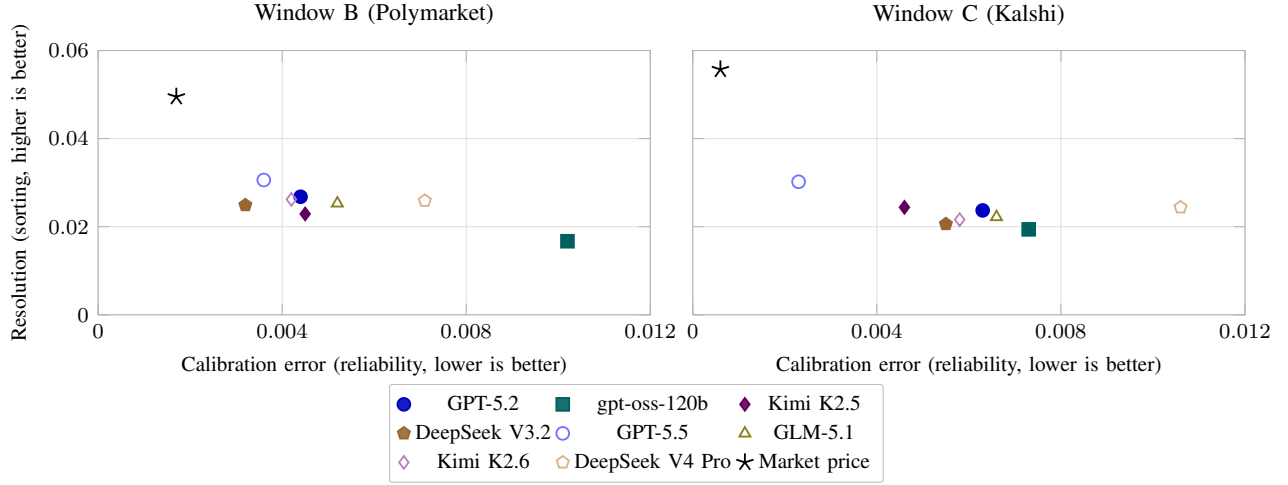


Fig. 3: The gap is sorting, not calibration. Each point is one forecaster’s calibration error (across, lower is better) against its resolution (up, higher is better). Ten probability bins; the bin-free decomposition gives the same picture. Test split of the primary set.

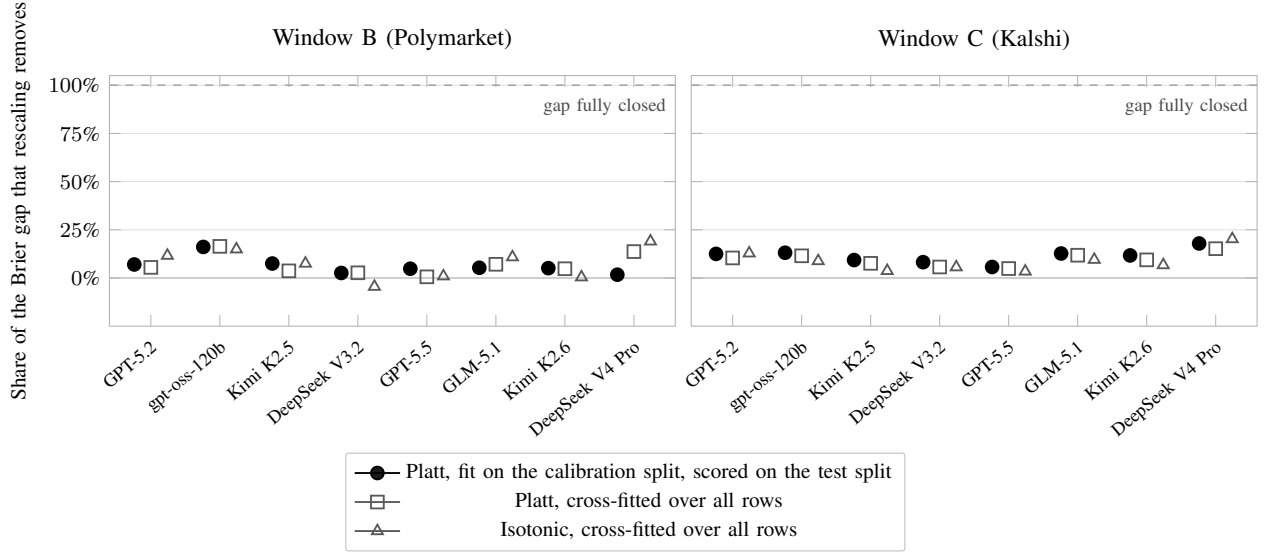


Fig. 4: Rescaling cannot close the gap. Each point is the share of one model’s Brier shortfall removed by an after-the-fact rescaling. Row sets differ by series: the single-split maps are fitted on the calibration split and scored on the test split, while the two cross-fitted series score every row. This is the one analysis not reported on the test split alone; cross-fitting admits no alternative. Window A is not shown: slopes above one, recovery between -9 and $+2$ per cent (single split) and between -14 and $+14$ per cent (cross-fitted).

TABLE II: Leaning with or against the crowd. Rows where the market sits at exactly 0.5, or the model exactly at the price, have no direction and are in neither group, so shares do not sum to 100%; they still count in the net deficit quoted in the text, which is why shares recomputed from this table alone differ slightly on Window A. Brier edge is the market’s score minus the model’s. All models pooled, test split of the primary set; brackets are 95% event-resampling ranges. Return averages each trade’s own return after fees, so it is not win share over price paid.

Window	Model leans	Rows	Share	Brier edge	Return per \$1	Price paid	Win share
A	With	2,236	45%	+0.005 [−0.003, +0.012]	−0.003 [−0.069, +0.053]	.710	.707
	Against	2,627	53%	−0.038 [−0.060, −0.017]	−0.172 [−0.309, −0.017]	.294	.267
B	With	4,536	38%	−0.003 [−0.010, +0.004]	+0.017 [−0.035, +0.069]	.695	.716
	Against	7,051	60%	−0.046 [−0.060, −0.033]	−0.060 [−0.173, +0.064]	.282	.288
C	With	5,322	31%	−0.011 [−0.018, −0.004]	−0.043 [−0.094, +0.001]	.712	.700
	Against	11,356	66%	−0.052 [−0.062, −0.042]	−0.124 [−0.218, −0.034]	.305	.287

A remaining explanation is shared missing information: the price carries facts no model was given, and forecasters missing the same facts miss the same way. On news-bearing rows asked with and without the news, adding it lowers the residual correlation by 0.047, 0.055 and 0.036, clear of zero on Window A and Window B, but not clearly through what the models share: on Window A each model's own spread widens while the shared part does not fall, and on Window B and Window C the shared part falls but not clearly (Appendix H). News cannot carry line-ups or injuries, and the design cannot separate what remains from shared training or the shared prompt, which we did not vary (Section 7). Forecasters given identical inputs would agree strongly even if each reasoned perfectly, so the result shows that the models are not independent opinions, not that they share judgment.

A second measure counts how often two models land on the same incorrect side of even money. On rows the market itself got right, pairs miss together 4.3, 4.0 and 2.7 times the independent rate, against a shrinkage null of 2.0 to 2.2, 1.8 to 1.9 and 1.4 to 1.5; nulls that also share each row's features, or the event, stay below the real ratio (Appendix G).

Same-developer pairs sit at both ends of the range, the tightest (Kimi K2.5 with Kimi K2.6, 0.910 on Window B) and the loosest (GPT-5.5 with gpt-oss-120b, 0.626 on Window C); with only three developers fielding more than one model we draw no conclusion (Appendix O).

That the models agree also decides what combining them can do. The arithmetic mean of all eight ties the best single model on Window B (Brier 0.2077 against 0.2076, both on the rows every model forecast, which is why they differ slightly from Table III) and is worse on Window C (0.2148 against 0.2117); the best single model is the same whether picked on the test rows or the calibration split. Learned weights help but not enough: non-negative least squares with an intercept, cross-fitted by event, reaches -0.0111 , -0.0208 and -0.0261 against the market on the same rows, an improvement of 0.0018, 0.0010 and 0.0012 on the best single model there, a small fraction of the width of that model's own interval. Every interval still lies below zero. The extremizing factor [23, 24] is 1.15 on Window A, 0.75 on Window B and 0.85 on Window C: on the two eight-model windows the data asks for the pooled forecast to be pulled toward even money rather than away from it.

6.5 Depth: no effect in returns, some in accuracy

If thin books hurt the models, losses should be largest where the book is thinnest. In returns they are not: the thin-minus-deep return gap is -0.018 on Window B and $+0.004$ on Window C, both intervals spanning zero by a wide margin, no individual model's gap clears zero, and regressing return on log depth gives no trend clear of zero. In accuracy the pattern is different. The models' Brier deficit shrinks as the book deepens: its slope on log depth is $+0.0075$ on Window B and $+0.0072$ on Window C, both intervals above zero, and the thin-minus-deep gap is -0.021 on Window B, its interval including zero, and -0.027 on Window C, clear of it. Holding price band or

horizon fixed keeps the slope clear of zero on Window C but not on Window B, and Window A shows no depth pattern in either measure. Returns are far noisier than Brier scores, so the two need not agree.

7 Limitations

These are ordered by how much they could move a conclusion.

1. The memorisation check could not be validated. Protection against a model recalling an outcome rests on the release-date rule and the memory probe. A model released long after Window A settled (GLM-5.3 Flash, 26 August 2026) was flagged at 1.4 per cent on 500 Window A rows, less than the 1.6 to 3.4 per cent for the eligible models on the same rows. The probe is therefore a filter of unknown power, and the leakage argument rests on release dates alone (Appendix X).

2. Short, sports-heavy windows. Sports make up 82, 73 and 81 per cent of primary rows, and our findings may not transfer to slower political or macroeconomic questions. The gap is larger on non-sports rows (Appendix P), so the sports concentration is not what produces it; but the non-sports evidence is thin.

3. One prompt, one sample, and what that means. Every model answered one prompt once, at temperature 0 where the host accepted it. We did not vary the prompt wording, so the measured agreement may overstate what would remain under varied prompts: we can say the models move together given a common question but cannot separate shared judgment from shared phrasing. The same-developer comparison does not settle it either way, for the reasons Section 6.4 gives. Nor can we say how much of each model's noise is specific to the one wording and draw it got; we did not average over prompt variants or repeated samples.

4. Thinking budgets are not comparable. At the 1,000-token cap sit 99 per cent of Kimi K2.5's and Kimi K2.6's rows, 86 to 92 per cent of DeepSeek V3.2's and 46 to 50 per cent of DeepSeek V4 Pro's (its median over the whole run, primary and non-primary rows together, sits at the cap in Table V), so we cannot say what they would do uncapped. The model-versus-market comparison does not depend on cross-model comparability, but a capped model might do better uncapped.

5. Hosts and numeric precision differ. Open-weight models are served at whatever precision the host offers, including int4 for Kimi K2.5, and one host truncated DeepSeek V4 Pro's output at 3,000 tokens. A differently served copy might score differently.

6. Question text on Window B and Window C was captured after settlement. Exchanges can edit text after listing, so equality with what a trader saw is assumed, not verified. Window A's text is verified, being captured live. Had an exchange clarified a rule mid-life, our models would have seen the clarified version and the crowd would not, which favours the models.

7. The news arm rests on few rows. Only 1,434, 563 and 752 primary rows carry news, and the analyses run on the test-split share of those.

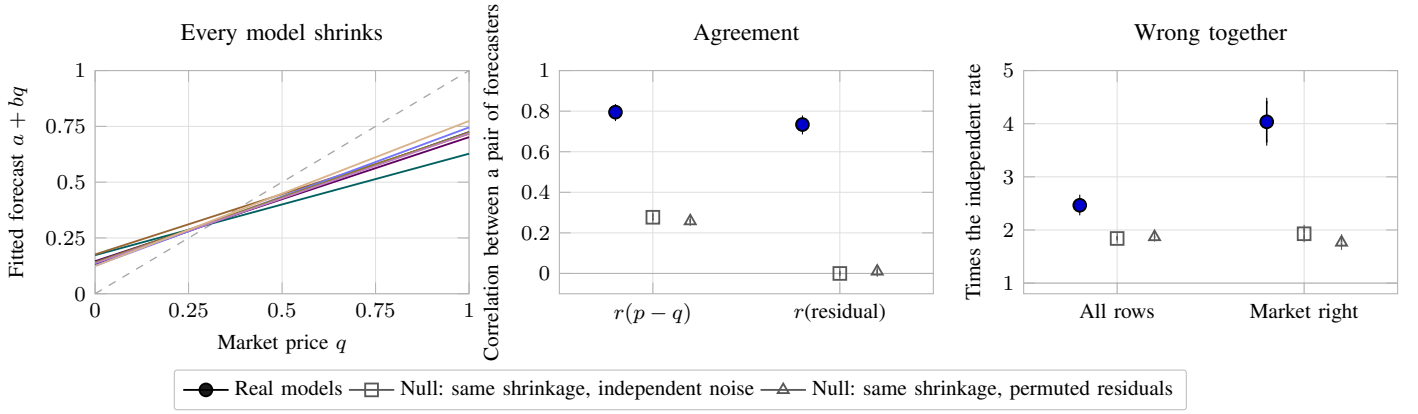


Fig. 5: Shared shrinkage does not account for the agreement. Left: fitted $p = a + bq$ for each model on Window B; every slope is below one. Middle and right: simulated forecasters carrying each model’s fitted a , b and residual spread but independent noise agree on departures at 0.26 to 0.28 on this window (0.23 to 0.28 across all three) and on residuals at essentially zero, by construction; the real models reach $r(p - q) = 0.795$, residual $r = 0.734$ and a wrong-together ratio of 2.47. Real values carry 95% event-bootstrap intervals (1,000 resamples, shrinkage refitted inside each); simulated ranges are 2.5th–97.5th percentiles over 400 draws. Test split, rows every model forecast.

8. The price is not a certified probability, and the midpoint is not executable. The accuracy comparison is against the midpoint of the best bid and ask, a standard summary but not a price anyone can trade at; the trading simulation fills at the quoted bid and ask with fees charged, and the two should not be read as the same exercise. The price bounds the mean belief only within an interval, wide except near even money [25], though in practice prices are usually close to mean beliefs [26]. Our claims are about the price as a forecast; a reader who wants “collective human belief beats the models” needs the extra assumption.

9. The prompt is price-blind. Our results say nothing about how a model would trade with the price in view.

10. Trading is simulated from a snapshot. There is no queue position, price impact, partial fill or later opportunity to trade, so returns are an optimistic bound. This matters most for the against-the-crowd rows where the losses are largest.

11. Window C cannot be fully reproduced from the release. Kalshi’s terms do not permit redistributing market data, so Window C ships as identifiers and model outputs only.

12. No non-LLM sports baseline. We have not run a simple non-LLM baseline on the sports rows, which would bound how much of the market’s advantage needs no language model.

8 Conclusion

On two venues, over three short windows of mostly sports markets, with live books required on two of them, models that never saw a price, and a roster restricted to versions published before the outcomes they forecast, the market was more accurate than every model we scored. The gap is sorting rather than calibration: resolution accounts for 79 to 99 per cent of it (bin-free), and refitting each model’s probability scale on held-out data leaves most of it standing, even on the non-sports rows where calibration matters most. Part of the gap is information, but news never closes it and the effect

is individually significant on only one window. The models err together beyond what their fitted lines to the price or a row’s observable features explain; supplying news lowers that agreement only slightly. In accuracy, though not in returns, the models’ deficit shrinks as books deepen, clearly on one window and less robustly on another.

Because it is a resolution gap, better calibration will help little outside non-sports rows; the models would have to discriminate better, not report better. And because the errors are shared, no combination of the models reached the market: equal-weight averaging ties or trails the best single model, the extremizing factor fitted on held-out data is below one on both eight-model windows, and cross-fitted learned weights gain at most about 0.002 in Brier over the best single model, still clearly short of the price. At these thinking budgets GPT-5.5 sorts best, yet clearly trails the market and errs with the rest.

We are cautious about markets that admit AI traders: our models never saw a price, so we cannot say how they would behave inside one. The correlated-error result establishes a precondition for that concern, not the concern itself: under this protocol the signals these models produce are common to them, whether from shared ignorance or shared judgment, so a market drawing on many of them may be drawing on fewer independent opinions than the count suggests [22].

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Appendix A

Absolute scores

The main text reports differences, because the level of a Brier score is set by how hard a window’s questions happened to be. Table III gives the levels themselves, so readers need not reconstruct them. Uncertainty (the score of a forecaster who answers the window’s base rate on every row) is a property of the outcomes, so it is the same for everyone in a window and appears in the heading. Skill is the share of that forecaster’s error removed.

The constant forecasters are included to make the scale legible, not because they are contenders. Answering Yes on every row scores worse than the base rate by a factor of two or more, and answering 0.5 on every row scores worse than the base rate on all three windows. Beating them is not evidence of judgment; the informative comparisons are with the market and the shrunk market.

The shrunk market is a linear, and so monotone, compression of the price. Applied to the price it is pure calibration loss, yet it reproduces 24 to 52 per cent of the models’ shortfall while calibration explains much less of it. The two fit together: a calibrated forecaster that knows less than the market also varies less than the price, and in that forecaster the compression is lost resolution, not miscalibration.

Appendix B

Which file produces which result

Every number in this paper comes from a released file. Table IV maps each float and each named analysis to the script or results file that produces it. Scripts (a name with no extension, or .py) and models.json are in the code repository, each exploratory script with its saved output; result tables (.csv) and the other .json inputs are in the main dataset. Window C numbers also need the Kalshi market data, which the dataset card explains how to obtain and build_db.py merges.

Appendix C

Fee formulas

On Polymarket the taker fee is a topic-dependent rate times $p(1 - p)$: sports 0.05 from July 2026, crypto 0.07, politics and tech 0.04, geopolitics free. On Kalshi the fee is 0.07 times a per-market multiplier times $P(1 - P)$, rounded up to the next cent per order.

Appendix D

News attachment rules

An article may attach to a row only if its crawl time is strictly before the forecast moment and no more than fourteen days before it; it must match the market by name phrase and clear a reranker score of 0.3; at most the three best-scoring articles attach. Two dateline rules then apply. The benchmark drops any row whose news carries a dateline two or more days after the news’s own capture time. At call time the harness separately blocks any row whose news carries a dateline two or more days after the forecast moment, unless that date was hand-labelled as a scheduled future event rather than a publication date: 22 rows in all (17 on Window A, 3 on Window B, 2 on Window C, at most 0.6 per cent of a window’s in-play rows), labelled from the article text alone, without access to prices, outcomes or model forecasts. One day is allowed in both because capture times are UTC and sites print local dates.

Appendix E

Skill score

The skill score against a forecaster who answers the window’s base rate $\bar{y}$ on every question is

$$S(p) = 1 - \frac{BS(p)}{BS(\bar{y})}, \quad (2)$$

where zero means no better than answering the base rate every time and one means perfect.

Appendix F

Correlated-error method

For model m we fit

$$p_i^{(m)} = a_m + b_m q_i + \varepsilon_i^{(m)} \quad (3)$$

by least squares on the probability scale. We call a slope $b_m < 1$ shrinkage: the model moves less than the price does, whether because it hedges or because it knows less, and its forecasts sit toward the fixed point $a_m/(1 - b_m)$. The quantity we report is the mean pairwise correlation of the fitted residuals,

$$\bar{\rho} = \left(\frac{M}{2}\right)^{-1} \sum_{m < m'} \text{corr}(\varepsilon^{(m)}, \varepsilon^{(m')}), \quad (4)$$

which is agreement that each model’s own fitted shrinkage cannot explain.

Two simulated controls appear in Figure 5. Both build forecasters carrying each model’s fitted a_m and b_m ; they differ in where the noise comes from. The first draws it from a Gaussian at the model’s own fitted residual spread. The second discards the distributional assumption and resamples each model’s *observed* residuals without replacement within a price band, so the simulated noise has the real shape and the real spread and only its pairing across models is destroyed. The two agree closely, which is why the text quotes them as a range. Neither is a demanding test: independent noise is uncorrelated by construction, so these controls show only that shrinkage alone cannot manufacture the agreement, not that nothing else could.

The stronger control replaces $b_m q_i$ with a fitted surface that includes the row’s observable characteristics (a sports indicator, topic tags, horizon in log days and as band indicators, a news indicator and price-band indicators) and, on rows belonging to events with more than one market, a separate intercept per event. Section 6.4 reports $\bar{\rho}$ under each fit.

The raw error correlation, $\text{corr}(p - y)$, is 0.9625 on Window B and 0.9599 on Window C on the test split. We report it because earlier work uses it [21], but it should not be compared with $\bar{\rho}$: a raw error correlation rises whenever forecasts move little relative to outcomes.

Appendix G

Wrong together: per-window detail

A pair lands on the same incorrect side of even money 26.4 per cent of the time on Window B against 10.7 per cent expected under independence, a ratio of 2.47, and 26.9 against 13.3 on Window C, a ratio of 2.03. On news-bearing rows the ratios are 2.33 and 2.56 (431 and 594 rows), read against a shrinkage null of 1.67 and 1.78 on those same rows; restricted further to the news-bearing rows the market got right, they are 3.11 on Window B and 4.10 on Window C against nulls of 1.4 to 1.8. Simulated forecasters sharing only shrinkage already miss together about 1.9 times as often as independence on all rows of Window A and Window B and about 1.5 times on Window C, so the margin is narrower than a comparison against independence alone would suggest. Between 58 and 74 per cent of shared mistakes fall where the market was also wrong, leaving a quarter to two fifths that are the models’ own.

Richer nulls do not close the gap. On rows the market got right, simulated forecasters that share each model’s fit to the price and the row’s observable features miss together 2.3 times the independent rate on Window A, 1.9 to 2.0 on Window B and 1.5 on Window C, against real ratios of 4.3, 4.0 and 2.7. Adding a separate intercept per event, on the rows of events with more than one market, raises the null to 3.5 to 3.6, 2.9 and 2.2 to 2.3, against real ratios on those rows of 5.9, 4.3 and 3.2. Each null as a share of the real ratio on its own rows

TABLE III: Absolute scores on the test split of the primary set. Brier and reliability are lower-is-better; resolution and skill are higher-is-better. Constant forecasters have no resolution by construction and their reliability is not comparable with a varying forecaster’s, so those cells are left blank. Reliability minus resolution plus uncertainty differs from Brier by up to 0.002, the within-bin term that ten bins leave out [17]. The market rows average the market’s score over every model’s rows stacked together, so they differ in the fourth decimal from the market score behind any one model’s Brier edge, which is computed row by row on that model’s own rows. The Brier edge (market minus model, on each model’s own rows) and the market-minus-model resolution carry 95% event-resampling ranges.

Forecaster	Rows	Brier	Brier edge	REL	RES	Market minus model RES	Skill
Window A (uncertainty 0.2317, Yes base rate 0.365)							
GPT-5.2	1,248	0.2007	−0.0153 [−0.0265, −0.0051]	0.0051	0.0352	+0.013 [+0.003, +0.024]	+0.134
gpt-oss-120b	1,248	0.2159	−0.0305 [−0.0449, −0.0162]	0.0074	0.0214	+0.027 [+0.015, +0.040]	+0.068
Kimi K2.5	1,237	0.1990	−0.0129 [−0.0264, −0.0013]	0.0033	0.0352	+0.013 [+0.002, +0.024]	+0.142
DeepSeek V3.2	1,248	0.1997	−0.0143 [−0.0262, −0.0030]	0.0023	0.0338	+0.015 [+0.003, +0.028]	+0.138
the market	—	0.1855	—	0.0015	0.0486	—	+0.199
Market, shrunk	—	0.1951	—	—	—	—	+0.158
Base rate every row	—	0.2317	—	—	0.0000	—	+0.000
0.5 every row	—	0.2500	—	—	0.0000	—	−0.079
Yes every row	—	0.6352	—	—	0.0000	—	−1.741
No every row	—	0.3648	—	—	0.0000	—	−0.574
Window B (uncertainty 0.2352, Yes base rate 0.378)							
GPT-5.2	1,480	0.2120	−0.0255 [−0.0344, −0.0169]	0.0044	0.0268	+0.023 [+0.015, +0.030]	+0.099
gpt-oss-120b	1,480	0.2281	−0.0416 [−0.0517, −0.0310]	0.0102	0.0167	+0.033 [+0.024, +0.041]	+0.030
Kimi K2.5	1,477	0.2158	−0.0294 [−0.0389, −0.0205]	0.0045	0.0229	+0.027 [+0.018, +0.035]	+0.082
DeepSeek V3.2	1,480	0.2138	−0.0272 [−0.0361, −0.0182]	0.0032	0.0249	+0.025 [+0.017, +0.032]	+0.091
GPT-5.5	1,480	0.2080	−0.0214 [−0.0301, −0.0128]	0.0036	0.0306	+0.019 [+0.011, +0.027]	+0.116
GLM-5.1	1,480	0.2149	−0.0284 [−0.0379, −0.0189]	0.0052	0.0253	+0.024 [+0.016, +0.033]	+0.086
Kimi K2.6	1,479	0.2126	−0.0264 [−0.0351, −0.0180]	0.0042	0.0262	+0.023 [+0.016, +0.031]	+0.096
DeepSeek V4 Pro	1,475	0.2160	−0.0298 [−0.0401, −0.0200]	0.0071	0.0259	+0.024 [+0.016, +0.032]	+0.082
the market	—	0.1864	—	0.0017	0.0500	—	+0.207
Market, shrunk	—	0.1956	—	—	—	—	+0.168
Base rate every row	—	0.2352	—	—	0.0000	—	+0.000
0.5 every row	—	0.2500	—	—	0.0000	—	−0.063
Yes every row	—	0.6217	—	—	0.0000	—	−1.643
No every row	—	0.3783	—	—	0.0000	—	−0.608
Window C (uncertainty 0.2402, Yes base rate 0.401)							
GPT-5.2	2,137	0.2218	−0.0372 [−0.0447, −0.0293]	0.0063	0.0237	+0.032 [+0.025, +0.038]	+0.077
gpt-oss-120b	2,137	0.2278	−0.0431 [−0.0515, −0.0347]	0.0073	0.0194	+0.036 [+0.029, +0.043]	+0.052
Kimi K2.5	2,134	0.2205	−0.0361 [−0.0441, −0.0285]	0.0046	0.0244	+0.031 [+0.024, +0.039]	+0.082
DeepSeek V3.2	2,137	0.2248	−0.0402 [−0.0480, −0.0325]	0.0055	0.0206	+0.035 [+0.028, +0.041]	+0.064
GPT-5.5	2,136	0.2117	−0.0272 [−0.0341, −0.0200]	0.0023	0.0302	+0.026 [+0.019, +0.032]	+0.119
GLM-5.1	2,137	0.2255	−0.0409 [−0.0489, −0.0330]	0.0066	0.0222	+0.034 [+0.027, +0.039]	+0.061
Kimi K2.6	2,136	0.2245	−0.0397 [−0.0477, −0.0320]	0.0058	0.0216	+0.034 [+0.027, +0.041]	+0.066
DeepSeek V4 Pro	2,130	0.2264	−0.0418 [−0.0506, −0.0334]	0.0106	0.0244	+0.031 [+0.024, +0.038]	+0.058
the market	—	0.1846	—	0.0006	0.0558	—	+0.232
Market, shrunk	—	0.1936	—	—	—	—	+0.194
Base rate every row	—	0.2402	—	—	0.0000	—	+0.000
0.5 every row	—	0.2500	—	—	0.0000	—	−0.041
Yes every row	—	0.5990	—	—	0.0000	—	−1.494
No every row	—	0.4010	—	—	0.0000	—	−0.669

is 55, 47 to 48 and 58 per cent for features (all market-right rows) and 60 to 61, 67 and 69 to 70 per cent with the event (multi-market rows), so the event appears to absorb part of the excess, though the two shares rest on different rows. The real ratio still lies above every simulated range.

Appendix H

Agreement with and without news

Every news-bearing row was asked twice, with the news and with it removed, so the models’ agreement can be measured on the same rows under both. The residual correlation after each model’s fitted line is 0.756 with news against 0.803 without on Window A, 0.707 against 0.763 on Window B and 0.711 against 0.747 on Window C; the falls of 0.047, 0.055 and 0.036 exclude zero on Window A and Window B and reach it on Window C (event-resampled, both passes resampled together, on the 1,141, 431 and 593 rows every model answered in both passes; on one Window C row a model has no answer without the news, hence 593 here against 594 in Appendix G).

The wrong-together ratio on rows the market got right shows no clear change: +0.14, −0.26 and +0.20, every interval spanning zero. News therefore removes a slice of the agreement in departures from the price and none detectable in shared wrong calls. The test bounds only the information our news carried.

A correlation can fall because the models share less or because each becomes noisier. Splitting the change settles which (same rows, same resampling). On Window A the residual standard deviation rises from 0.154 to 0.168 with news, clear of zero, while the mean pairwise covariance rises slightly: holding the spreads at their no-news values, the with-news correlation would be 0.90, not 0.76. The fall on Window A is therefore model-specific noise added by the news, on the one window whose news was retrieved at run time. On Window B and Window C the spreads barely move (−0.002) and the covariance falls, by 0.0026 and 0.0020, which alone would lower the correlation by 0.07 and 0.06; neither covariance change clears zero. The shared-missing-information reading is therefore suggestive on the two live-book windows and unsupported on Window A.

TABLE IV: Where each result comes from.

Result	Produced by
Table I	benchmark-primary-set-counts, depth_cuts.json, reporting_line.csv, expB_main.csv
Table II	direction-and-money-against-crowd-carries-the-loss, direction-and-money-against-means-buying-the-longshot
Table III	baselines-absolute-and-horizon, calibration-shape-resolution-gap, baselines-whole-window-edge
Table V	models.json, reporting_line.csv
Figure 1 and the Yes-side cost (Section 6.3)	news-effect-high-price-gap, calibration-shape-yes-no-asymmetry (test-split output)
Figure 2	expA_buckets.csv
Figure 3	calibration-shape-resolution-gap
Figure 4	recalibration-crossfit-and-isotonic
Figure 5	correlated-errors-shrinkage-null
Figure 6	correlated-errors-pair-ranking, expC_overall.csv
Shared inputs control (Section 6.4)	correlated-errors-covariate-adjusted
Agreement with and without news (Appendix H)	correlated-errors-with-and-without-news, correlated-errors-news-covariance-or-variance
Depth in accuracy terms (Section 6.5)	expA_gap_trend.csv, expA_pooled_verdicts.csv
Learned combination (Section 6.4)	ensembles-learned-stacking, ensembles-and-extremizing
Bin-free decomposition (Section 6.2)	decomposition-binning-bias
Failure imputation and re-ask rate (Section 4)	failures-imputation-sensitivity, probe-news-value-and-reask-rate
Forecast horizon (Appendix T)	baselines-absolute-and-horizon
Fee and stake sensitivity (Appendix N)	fees-stake-sensitivity
News placebo (Section 6.3)	news-toward-away-placebo
Memory probe and value of news (Appendix X, Section 6.3)	quarantine_<run>-summary.json, window_c_quarantine_counts.csv, first_runs_report.py, probe-news-value-and-reask-rate
Skill scores and shrunk market (Section 6.1)	baselines-absolute-and-horizon
Non-ladder in-play rows and ladders (Section 6.1)	secondary_full_set.csv, secondary_ladders.csv
Wrong together (Appendix G)	correlated-errors-shrinkage-null, correlated-errors-shrinkage-null-news-rows, correlated-errors-covariate-adjusted, expC_overall.csv
Newest model (Appendix I)	pair_gpt-5.2_vs_gpt-5.5_paired.csv, calibration-shape-model-spread, decomposition-binning-bias
Same side or opposite (Appendix J)	direction-same-side-share, direction-and-money-against-means-buying-the-longshot
Events and clusters (Appendix K)	event-definition-and-cluster-counts
Split leakage (Appendix L)	split-leakage-event-grouped
Small reliability term (Appendix M)	decomposition-why-reliability-is-small
Pairwise correlations (Appendix O)	correlated-errors-pair-ranking
Non-sports gap (Appendix P)	topic-and-sports-brier-gap (test-split output), topic-and-sports-divergence, calibration-shape-sports-split
Thinking length (Appendix Q)	thinking-and-length-more-thinking-more-disagreement, thinking-and-length-budget-cap-harmless
Calibration gate (Appendix R)	calibration_gate.json, gate_used.json, expB_main.csv
Convergence (Appendix S)	expB_main.csv, convergence-toward-outcome, convergence-settled-in-window, convergence-baseline-choice, convergence-open-market-move
Exploratory scripts (Appendix U)	every script in analysis/exploratory/, each with its saved output in analysis/exploratory/outputs/
Prompt (Appendix W)	prompt.py

Appendix I

The two newest models

GPT-5.5 has the highest resolution in the study, 0.0306 on Window B and 0.0302 on Window C, and is still short of the market by +0.019 and +0.026 with both intervals clear of zero. Against GPT-5.2 it is better on Kalshi by 0.0099 (interval excluding zero) and not clearly better on Polymarket. DeepSeek V4 Pro shares the same release date yet on Window C has a resolution gap to the market of +0.0314 against GPT-5.5's +0.0257, so recency does not separate models released on the same day. DeepSeek V4 Pro is a capped model whose host truncated some outputs, so one pair is thin evidence either way.

Appendix J

Leaning against the crowd: where the loss sits

Most rows that lean against the crowd are same-side-less-confident rather than outright contradiction, but the accuracy loss is not spread evenly across the two. On the rows where the model genuinely favours

the other outcome the Brier edge is -0.084 on Window A, -0.121 on Window B and -0.104 on Window C, against -0.016 , -0.013 and -0.019 on the same-side rows: five to ten times worse. The direction split therefore tracks something real, even though the majority of the category is a difference of confidence. In money the contrast is the same shape: trades placed against the crowd buy at 0.28 to 0.31 and win 27 to 29 per cent of the time, while trades placed with the crowd pay 0.70 to 0.71 and win 70 to 72 per cent. These averages look close to fair, but returns average each trade's own payoff, and on Window A, where fees were near zero, against-the-crowd trades still lost 17 cents per dollar.

In the lowest price band the models answer 0.20 to 0.25 against the crowd's 0.13 to 0.15. Each model's fixed point sits near the window's base rate: 0.26 to 0.34 on Window A (base rate 0.37), 0.32 to 0.38 on Window B (0.38) and 0.38 to 0.44 on Window C (0.40).

Appendix K

What an event is, and how many clusters

Every interval in this paper resamples *events* rather than rows. An event is the venue's own grouping of markets about one happening (a Polymarket event id, a Kalshi event ticker), so one event may carry nineteen markets on which songs are played at a single halftime show, or twelve on what one official says at one press conference. Markets inside an event move together, so treating them as independent draws would give intervals that are too narrow.

On the test split the bootstrap draws from 626 events on Window A, 1,196 on Window B and 1,823 on Window C, holding 1,248, 1,480 and 2,137 market-moments. Markets per event in the test split run from one to thirteen. How much the clustering matters depends on the window: 46.6 per cent of Window A events carry a single market, against 83.5 per cent on Window B and 89.1 per cent on Window C. On the two larger windows the event bootstrap is therefore close to a row bootstrap, and the protection it offers is real but small; on Window A it does substantial work.

Appendix L

Does the calibration split leak across events?

The calibration and test split was drawn per market rather than per event, so two markets from one event can land on opposite sides. Since the recalibration control fits a map on one side and scores it on the other, a map might be fitted on rows closely related to those it is scored on. The exposure is real: straddling events hold 31.3 per cent of Window A test rows, 15.7 per cent of Window B and 10.2 per cent of Window C.

We therefore redrew the split, same cells and same size, keeping every event whole, and refitted the control. Recovery does not rise materially. On Window B it runs from -0.5 to 17.1 per cent against 1.7 to 16.1 per cent under the main split; on Window C every model recovers *less*, the range falling from 5.7 to 17.9 per cent down to 0.6 to 9.9 per cent. Scoring the main-split map on non-straddling rows alone moves nothing systematically. The cross-fitted Platt and isotonic figures, which set the paper's ceiling of about a fifth, are unaffected either way because those folds are built by event. The conclusion that no monotone rescaling closes the gap is unchanged, and on Window C it is stronger.

Appendix M

Why the reliability term is small

Reliability being small does not mean the models are well calibrated, and three separate things keep it small. First, it is a squared quantity: a per-model reliability of 0.0023 to 0.0106 is a root-mean-square miss of roughly 0.05 to 0.10 in probability, set against a market-minus-model resolution gap of 0.019 to 0.036 . Second, the badly calibrated forecasts sit in the thin tails. About three in ten forecasts fall outside 0.2 to 0.7 , and those carry 68 per cent of the total reliability on Window A, 95 per cent on Window B and 78 per cent on Window C, while the crowded middle bins miss by only 0.012 to 0.039 . Third, ten equal-width bins hide disagreement inside a bin. Splitting each bin by whether the market priced the row above or below the bin's mean forecast multiplies the pooled reliability by three to five, from 0.0027 to 0.0128 on Window A, 0.0043 to 0.0130 on Window B and 0.0055 to 0.0191 on Window C; a random split into halves of the same size leaves it almost unchanged. That third figure is a diagnostic, not a competing decomposition: Murphy's reliability conditions on the forecast alone, so conditioning on the price as well moves part of the sorting shortfall into the calibration column by construction. Measured against a model's own information the shortfall is a sorting gap; measured against the market's better information the same fact appears as miscalibration. The 79 to 99 per cent attribution in Section 6.2 is the former. Note that the market sits in the 0.2 to

0.7 range about as often as the models do, with reliability three to nine times smaller, so the range alone explains nothing.

Appendix N

Fee and stake sensitivity

The same trades with no fee at all return -0.008 on Window B against -0.033 with fees, and -0.059 on Window C against -0.099 : before costs the Window B trades are roughly break-even. On Window A fees were about 0.02 per cent of outlay, so the return of -0.102 is the same with or without them. Raising the stake from \$1 to \$1,000 moves the Kalshi return by 0.4 percentage points, so the result is not an artefact of small orders.

Appendix O

Pairwise departure correlations

Among the five same-developer pairs, two are the tightest in the roster: Kimi K2.5 with Kimi K2.6 at 0.910 on Window B and 0.891 on Window C, and GPT-5.2 with GPT-5.5 at 0.872 on Window B, which is what a same-developer story predicts. The two DeepSeek models are no closer than unrelated pairs, sitting 0.061 and 0.026 below the roster reference; gpt-oss-120b's pairs sit 0.088, 0.099 and 0.090 below it and are six of the seven loosest on Window B. On two windows of three the loosest pair is itself same-developer: GPT-5.2 with gpt-oss-120b at 0.722 on Window A and GPT-5.5 with gpt-oss-120b at 0.626 on Window C.

Appendix P

The non-sports gap

Sports dominate all three windows, so it matters that the model-to-market gap is not a sports artefact. It is not: on the non-sports minority the gap is several times larger. Pooling the roster, the Brier edge on sports rows is -0.0061 , -0.0131 and -0.0266 on Window A, Window B and Window C, against -0.0739 , -0.0727 and -0.0873 on non-sports rows. The sports-minus-non-sports difference is $+0.068$, $+0.060$ and $+0.061$, with every interval clear of zero, and each model's own difference clears zero on every window. Per model the ratio of the two gaps runs from 2.4 to 6.7 on Window B and Window C. On Window A the ratios are 4.9, 11.7 and 34.0 and one is undefined, because the sports gap there is so near zero (Kimi K2.5's is slightly positive) that the ratio stops being meaningful; the difference, not the ratio, is the stable quantity on Window A. The practical reading is that the sports concentration described in Section 7 makes the models look better than they are, not worse.

Sorting and calibration hold up within each half. Run separately on the test split's sports and non-sports rows, the bin-free decomposition puts 85 to 109 per cent of the shortfall on resolution for sports rows on Window B and 87 to 96 per cent on Window C (above 100 where the model is better calibrated than the market), and cross-fitted rescaling recovers at most 18 and 11 per cent. On non-sports rows resolution still carries most of it, 73 to 90 per cent on Window B and 59 to 89 per cent on Window C, but rescaling recovers more: up to 30 and 42 per cent, largest for DeepSeek V4 Pro and, on Window C, GLM-5.1, the worst-calibrated models on those rows. On Window A the sports gap is so small (-0.001 to $+0.018$) that the shares are unstable; on its non-sports rows resolution carries 86 to 97 per cent and rescaling recovers at most 20. Calibration therefore matters more where the gap is widest, but it is not what the gap is made of.

Appendix Q

Thinking length within a model

Comparing thinking budgets *between* models is confounded (Section 7); comparing a model with itself removes that confound but not the difficulty of the rows. Splitting each model's rows into its

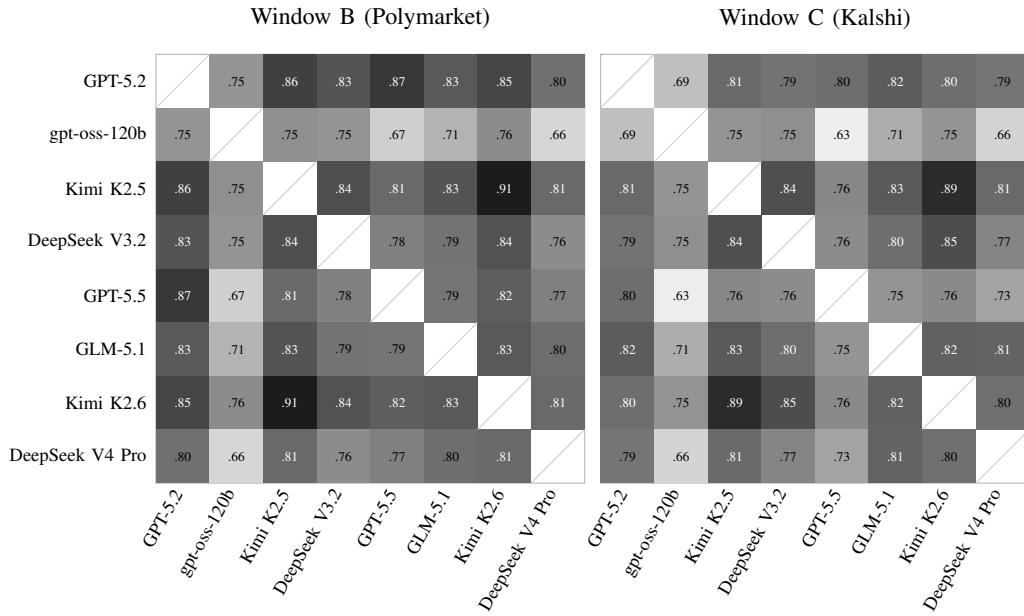


Fig. 6: Pairwise correlation of departures from the market price, $p - q$, on the rows all eight forecast. The mean across pairs is 0.78 to 0.80, against 0.23 to 0.28 for simulated forecasters sharing only shrinkage; after removing shrinkage the residual correlation is 0.71 to 0.75 against essentially zero, which is zero by construction (Appendix F). Test split of the primary set.

own top and bottom thinking quartile, the rows on which a model thought longest are those on which it departed further from the market and scored worse, not better. For GPT-5.2 the departure $|p - q|$ rises from 0.071 to 0.159 between bottom and top quartile on Window A, with the Brier edge falling from -0.0018 to -0.0230 ; the pattern repeats on Window B. GPT-5.5 behaves the same way, with departures $+0.085$ (Window B) and $+0.041$ (Window C) higher in its top quartile and Brier edges of -0.0372 and -0.0462 there, against -0.0049 and -0.0201 in its bottom quartile. This comparison uses both splits. Longer deliberation is associated with more disagreement with the crowd, and in this study disagreement with the crowd is what costs. We read this as selection rather than causation: hard rows invite more thinking.

Appendix R

The calibration gate

A disagreement with the price is only informative where a forecaster's probabilities mean something, so each model-window-bucket cell is admitted to the convergence test only if it holds at least 30 rows and its expected calibration error, computed over at most three bins of near-equal count, is at most 0.10; a cell with fewer than 30 rows stays closed. The gate is applied per cell rather than per model, so a model may enter the test in the buckets where it is calibrated and not in others. Failures are not concentrated in one model: in most failing cells the sports rows pass on their own and the non-sports rows are badly miscalibrated, the same asymmetry as Appendix P. Per-cell verdicts and admitted-row counts are in the released results tables.

Appendix S

Convergence: do prices drift toward the model?

A separate question is whether, when a model disagrees with the price, the price later moves toward the model. Measuring the share of disagreements whose subsequent price path drifts model-ward, and comparing it with the same statistic after shuffling which model disagreed on which row, the answer depends on the window and does not replicate. On Window A the drift exceeds the shuffled baseline for

all four models, by $+0.095$ to $+0.111$. On Window B the difference runs from -0.039 to $+0.042$ across the eight models, spanning zero, and on Window C it runs from -0.048 to -0.001 , that is, slightly the wrong way. We therefore make no claim that prices move toward these models. The Window A result is the one that would need explaining, and Window A is the window drawn without the live-book rule.

Appendix T

Forecast horizon

The size of the deficit depends strongly on how far ahead the forecast sits; its direction does not replicate across windows. Splitting rows by time to settlement, the pooled Brier edge varies by a factor of two to four within each window. On Window B the worst band is beyond 7 days (-0.063 against -0.016 at 3 to 7 days), and this is not an averaging artefact: beyond 7 days is the worst band for all eight models. On Window C the worst band is under a day (-0.052), again for all eight models. On Window A the worst is 1 to 3 days (-0.041). So horizon matters within a window and the models agree about how, but which end of the horizon hurts reverses between Window B and Window C, and we cannot say why.

One band is worth naming because it is the exception in the study. On Window A under a day the pooled edge is $+0.0229$, with an interval that crosses zero; for one model, Kimi K2.5, it is $+0.0452$ with an interval above zero. That is the only model-and-band cell anywhere in this work where a model is clearly more accurate than the price. It rests on 86 rows, on Window A, where only four models are eligible and on which we rest no claim (Section 6.1), so we report it rather than build on it.

Appendix U

Exploratory scripts

Every exploratory script in the release was run over all eight models and is released with its saved output; the appendix sections above cite the ones they use.

Appendix V

Roster, hosts, precision, thinking, failures and cost

Table V lists the eight scored models with the settings each was run under. Eligibility is by release date, as Section 4 describes: the four models published between August 2025 and January 2026 precede all three windows, while the four published in April 2026 precede Window C and Window B only. The validation model used in the memory probe (Appendix X) is never scored and is not listed. Across the whole study, 97,465 model calls were made, at a cost of about \$620.

Appendix W

The prompt

Figure 7 gives the template every scored call used, with the placeholders in braces filled per row. There is no price, bid, ask, depth or book anywhere in it. When a row has no news, the news block is the single line `No news is available`. A reply containing no parseable probability is re-asked once with the strict suffix shown beneath the template.

Appendix X

The memory probe

Every row carrying news was asked twice, once with the news and once with it removed. The pair serves two purposes.

Flagging memorisation. A model that answers a news-bearing question just as confidently and just as correctly without the news may be recalling the outcome rather than reasoning from what it was given. Rows meeting that condition are removed from the convergence test of Appendix S, the only analysis the quarantine touches. Across the 21 probe runs (the 20 scored-model runs and the validation run) the flag rate is 0.77 to 2.32 per cent.

Measuring what news is worth. The same pairing gives the value of news directly, as the Brier score without it minus the Brier score with it. On the test split it ranges from -0.001 to $+0.017$ by model and window and is positive in 18 of the 20 model-window cells, but small beside the gap to the market.

Validation, and why it is inconclusive. To check that the probe detects anything, a model published well after Window A settled (GLM-5.3 Flash, 26 August 2026) was run over 500 fixed Window A rows. If the probe worked, that model should have been flagged more often than the eligible models on the same rows. It was flagged less often: 1.4 per cent, seven rows of 500, against 2.2 per cent for GPT-5.2, 1.6 for DeepSeek V3.2, 2.2 for Kimi K2.5 and 3.4 for gpt-oss-120b. Either the probe lacks power, or that model had not memorised those particular markets. The exercise cannot separate the two, and we do not claim the probe as a working leakage detector; the release-date rule is what the leakage argument rests on.

TABLE V: The eight scored models. *Windows* lists the windows each model is eligible for. *Thinking* is either a reasoning-effort level (the OpenAI models and gpt-oss-120b) or a 1,000-token thinking budget; median thinking is tokens per reply, over every recorded reply of the run. The reply limit is 4,000 tokens for every model, although DeepSeek V4 Pro’s host truncated output at 3,000 on about 6% of its calls. *Failed rows* gave no usable probability after one stricter re-ask; counts are for the scored pass, and the memory-probe pass failed on a further 3 rows for Kimi K2.5, 7 for DeepSeek V4 Pro and 1 for Kimi K2.6. A dash marks a window a model was not eligible for, or a precision its host does not state. Sources: `models.json`, `reporting_line.csv`.

Model	Windows	Released	Host	Precision	Thinking	Median thinking A / B / C	Failed rows (all in play) A / B / C
GPT-5.2	A, B, C	2025-12-10	OpenAI	–	effort medium	233 / 198 / 223	0 / 0 / 0
gpt-oss-120b	A, B, C	2025-08-05	AkashML	bf16	effort medium	264 / 218 / 219	0 / 0 / 0
Kimi K2.5	A, B, C	2026-01-27	SiliconFlow	int4	cap 1,000 tokens	1,000 / 1,000 / 1,000	28 / 7 / 6
DeepSeek V3.2	A, B, C	2025-12-01	SiliconFlow	fp8	cap 1,000 tokens	1,000 / 1,000 / 1,000	0 / 0 / 0
GPT-5.5	B, C	2026-04-24	OpenAI	–	effort medium	– / 512 / 512	– / 0 / 7
GLM-5.1	B, C	2026-04-07	SiliconFlow	fp8	cap 1,000 tokens	– / 449 / 411	– / 0 / 0
Kimi K2.6	B, C	2026-04-20	SiliconFlow	fp8	cap 1,000 tokens	– / 1,000 / 1,000	– / 1 / 4
DeepSeek V4 Pro	B, C	2026-04-24	Alibaba	fp8	cap 1,000 tokens	– / 1,000 / 1,000	– / 9 / 17

You are forecasting the outcome of a prediction-market question.

Today’s date and time (UTC): {today}
Everything below is what is known as of today. Do not use or assume
knowledge of anything that happened after today.

Event: {event_title}
Question: {question}{contract_line}

Resolution rules (word for word from the exchange):
{description}

Scheduled market close: {market_end_date}

News:
{news_block}

Give your own probability that this question resolves Yes. Reason
briefly, then end your reply with one line in exactly this form:
PROBABILITY: <number between 0 and 1>

Your previous reply did not contain a usable probability. You must
answer with a number even if you are uncertain; refusing is not an
option. Reply with exactly one line and nothing else:
PROBABILITY: <decimal number between 0 and 1, for example 0.35>

Fig. 7: The forecasting prompt (above) and the strict re-ask appended after an unparseable reply (below). Braces mark per-row fields. Line breaks inside the running sentences are added here for column width; the template itself does not wrap them. Source: `harness/prompt.py`.